

# 3F7: Information Theory and Coding

## Handout 5: Relative Entropy, Mutual Information

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## Relative entropy

The *relative entropy* or the Kullback-Leibler (KL) divergence between two pmfs  $P$  and  $Q$  is

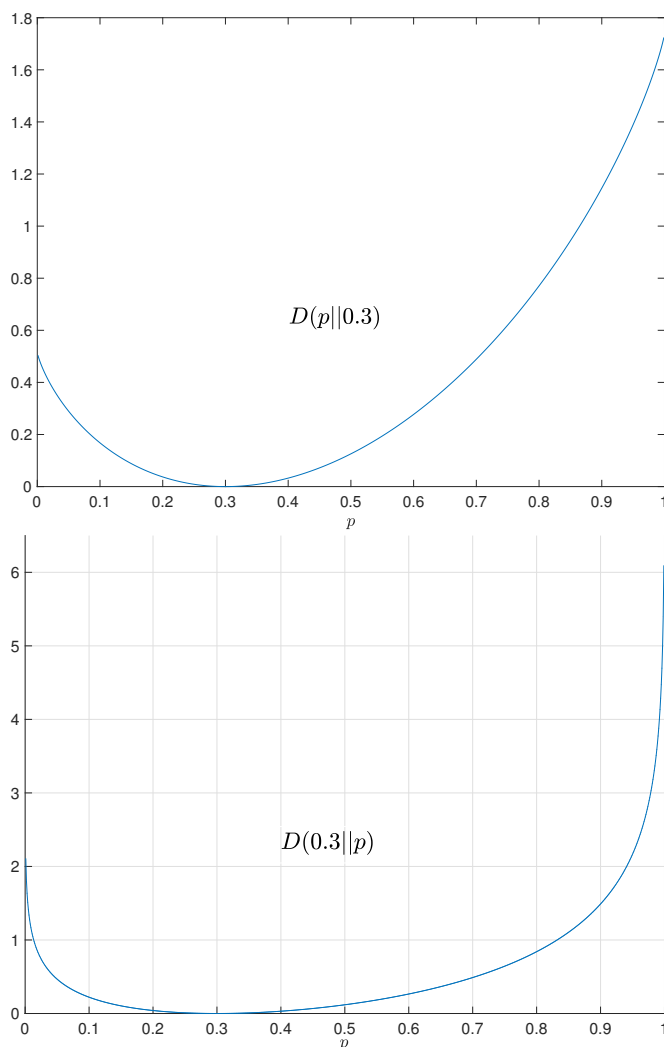
$$D(P\|Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

(Note:  $P$  and  $Q$  are defined on the same alphabet  $\mathcal{X}$ )

- ▶ Measure of distance between distributions  $P$  and  $Q$
- ▶ Not a true distance. For e.g.,  $D(P\|Q) \neq D(Q\|P)$ .

Example: If  $P = \text{Bern}(p)$  and  $Q = \text{Bern}(q)$  for  $p, q \in [0, 1]$ ,

$$D(P\|Q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$



Note the difference between  $D(\text{Bern}(p)||\text{Bern}(0.3))$  and  $D(\text{Bern}(0.3)||\text{Bern}(p))$

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Relative Entropy is always non-negative:  
 $D(P||Q) \geq 0$  with equality if and only if  $P = Q$ .

*Proof* : Using  $\log a = \frac{\ln a}{\ln 2}$ , we have

$$\begin{aligned} -D(P||Q) &= \frac{1}{\ln 2} \sum_{x \in \mathcal{X}} P(x) \ln \frac{Q(x)}{P(x)} \\ &\leq \frac{1}{\ln 2} \sum_{x \in \mathcal{X}} P(x) \left( \frac{Q(x)}{P(x)} - 1 \right) = 0, \end{aligned}$$

where inequality (a) uses  $\ln x \leq (x - 1)$  with equality iff  $x = 1$ . □

Relative entropy has many applications in information theory, statistics, and inference. We will look at two examples — one in [compression](#), and the other in [hypothesis testing](#).

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## Redundancy in Source Coding

Often the true distribution of the source is unknown, and we have to work with an estimated source distribution for compression.

- ▶ Suppose that the *true* pmf of a rv is  $P = \{p_1, \dots, p_m\}$ , and the estimated pmf is  $\hat{P} = \{\hat{p}_1, \dots, \hat{p}_m\}$ .
- ▶ We use  $\hat{P}$  to design a compression code whose code lengths  $\ell_i$  satisfy

$$\ell_i = \log(1/\hat{p}_i) \quad \text{for } i = 1, \dots, m.$$

In other words, the code is optimal for the distribution  $\hat{P}$ .

How far away are we from the optimal that could be achieved by knowing the true distribution? The average code-length is

$$\begin{aligned} L &= \sum_i p_i \ell_i = \sum_i p_i \log \frac{1}{\hat{p}_i} \\ &= \sum_i p_i \log \frac{p_i}{p_i \hat{p}_i} = \sum_i p_i \log \frac{1}{p_i} + \sum_i p_i \log \frac{p_i}{\hat{p}_i} \\ &= H(P) + D(P \parallel \hat{P}) \quad \text{bits/symbol.} \end{aligned}$$

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Therefore  $D(P \parallel \hat{P})$  is the price you pay in bits/symbol — *the redundancy* — for designing the code using  $\hat{P}$  rather than the true distribution  $P$ .

Suppose that we know that the true distribution  $P$  is one from a set of distributions  $\mathcal{P}$ . E.g., a ternary source that came from the set of distributions of the form  $\mathcal{P} = \{\gamma, 0.2, 0.8 - \gamma\}$  for  $\gamma \in (0, 0.8)$ .

Then we might like to design a code using a distribution  $\hat{P}$  that minimizes the *worst-case* redundancy over this class of distributions  $\mathcal{P}$ , i.e.,

$$\text{Choose } \hat{P} \text{ to minimize } \max_{P \in \mathcal{P}} D(P \parallel \hat{P}).$$

With this choice,

$$R^* = \min_{\hat{P}} \max_{P \in \mathcal{P}} D(P \parallel \hat{P})$$

is called the *minimax* redundancy.

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# Hypothesis Testing

In a basic version of simple hypothesis testing, we have data  $X_1, \dots, X_n$ , and the knowledge that *one* of the following is true.

$$H_0 : X_1, \dots, X_n \sim \text{i.i.d. } P$$

$$H_1 : X_1, \dots, X_n \sim \text{i.i.d. } Q$$

Based on the data  $X_1, \dots, X_n$  we have to decide whether hypothesis  $H_0$  or  $H_1$  is true.

## Examples:

1.  $P = \mathcal{N}(0, 1)$ , and  $Q = \mathcal{N}(\mu, 1)$  where  $\mu$  is the mean of a signal of interest. Thus, from noisy data  $X_1, \dots, X_n$ , we want to detect whether or not a signal is present.

2.  $P = \text{Bern}(p)$ , and  $Q = \text{Bern}(q)$ . Given the binary sequence  $X_1, \dots, X_n$ , we wish to determine whether it was generated from  $P$  or from  $Q$ .

$H_0$  is often called the null hypothesis

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## Some definitions

- ▶ A decision rule (or a test) is a function that maps the data  $(X_1, \dots, X_n)$  to a binary decision (0 for  $H_0$ , 1 for  $H_1$ ).
- ▶ With any decision rule, two kinds of errors can be made:
  - Type I error:** This occurs when  $H_0$  is true, but the decision rule chooses  $H_1$ .
  - Type II error:** This occurs when  $H_1$  is true, but the decision rule chooses  $H_0$ .
- ▶ Given the data  $X_1, \dots, X_n$ , the *likelihood ratio* (LR) is defined as

$$\frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)}.$$

and the normalized *log-likelihood ratio* (LLR) is

$$\frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)}$$

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For the problem of testing between distributions  $P$  and  $Q$ , the *optimal* decision rule is a likelihood-ratio thresholding rule, i.e.,

For some threshold  $T$ :

Choose  $H_1$  if  $\frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)} \geq T$ ; otherwise choose  $H_0$

Equivalently, the optimal rule can be expressed using LLR:

Choose  $H_1$  if  $\frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)} \geq t$ ; otherwise choose  $H_0$

(Note that both tests are equivalent by choosing  $T = 2^{nt}$ .)

- Observe that as we increase the threshold  $t$ , the condition to choose  $H_1$  becomes more stringent; hence type-I error prob. decreases and type-II error prob. increases.
- Conversely, as we decrease  $t$ , type-I error prob. increases and type-II error prob. decreases.
- The optimality of thresholding tests is in the following sense: **If a thresholding test has probability of type-I error  $\alpha$  and prob. of type-II error  $\beta$ , then any other test that has type-I error prob.  $\leq \alpha$  necessarily has type-II error prob.  $\geq \beta$ .**

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We will not prove optimality of thresholding tests here — this is called the “Neyman-Pearson lemma” [Cover & Thomas, Sec. 11.7]

**Example:** Under  $H_0 : X_1, \dots, X_n \sim \text{i.i.d. Bern}(p)$ ; under  $H_1$ , i.i.d.  $\text{Bern}(q)$

Denoting the number of ones in  $X_1, \dots, X_n$  by  $k_n$ , we have

$$\begin{aligned} \text{LLR}(X_1, \dots, X_n) &= \frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)} = \frac{1}{n} \log \frac{q^{k_n} (1-q)^{n-k_n}}{p^{k_n} (1-p)^{n-k_n}} \\ &= \frac{k_n}{n} \log \frac{q}{p} + \left(1 - \frac{k_n}{n}\right) \log \frac{1-q}{1-p} \end{aligned}$$

Q: What is the behaviour of the LLR as  $n$  gets large when: a)  $H_0$  is true; b)  $H_1$  is true.

For general  $P$  and  $Q$ , you will show in Examples Paper 2 that the normalized log-likelihood ratio

$$LLR(X_1, \dots, X_n) = \frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)}$$

satisfies the following:

1. Under  $H_1$ , where  $X_i \sim \text{i.i.d. } Q$ :

$$\frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)} \rightarrow D(Q \| P) \text{ in probability.}$$

2. Under  $H_0$ ,  $X_i \sim \text{i.i.d. } P$ , therefore

$$\frac{1}{n} \log \frac{Q(X_1, \dots, X_n)}{P(X_1, \dots, X_n)} \rightarrow -D(P \| Q) \text{ in probability.}$$

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Summarizing:

1. When the true distribution is  $Q$ , the likelihood of the data under the  $P$  distribution (for large  $n$ ) is

$$P(X_1, \dots, X_n) \approx Q(X_1, \dots, X_n) 2^{-nD(Q \| P)}$$

2. When the true distribution is  $P$ , the likelihood of the data under the  $Q$  distribution (for large  $n$ ) is

$$Q(X_1, \dots, X_n) \approx P(X_1, \dots, X_n) 2^{-nD(P \| Q)}$$

- ▶ In either case, the likelihood of the data under the wrong distribution is exponentially smaller than the likelihood under the true distribution.
- ▶ The exponent is controlled by the relative entropy, but note the asymmetry between  $D(P \| Q)$  and  $D(Q \| P)$ .

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# Mutual Information

Consider two random variables  $X$  and  $Y$  with joint pmf  $P_{XY}$ . The *mutual information* between  $X$  and  $Y$  is defined as

$$I(X; Y) = H(X) - H(X|Y) \quad \text{bits.}$$

*“Reduction in the uncertainty of  $X$  when you observe  $Y$ ”*

## Property 1

$$\begin{aligned} I(X; Y) &= H(X) + H(Y) - H(X, Y) \\ &= H(Y) - H(Y|X) \end{aligned}$$

*“ $X$  says as much about  $Y$  as  $Y$  says about  $X$ ”*

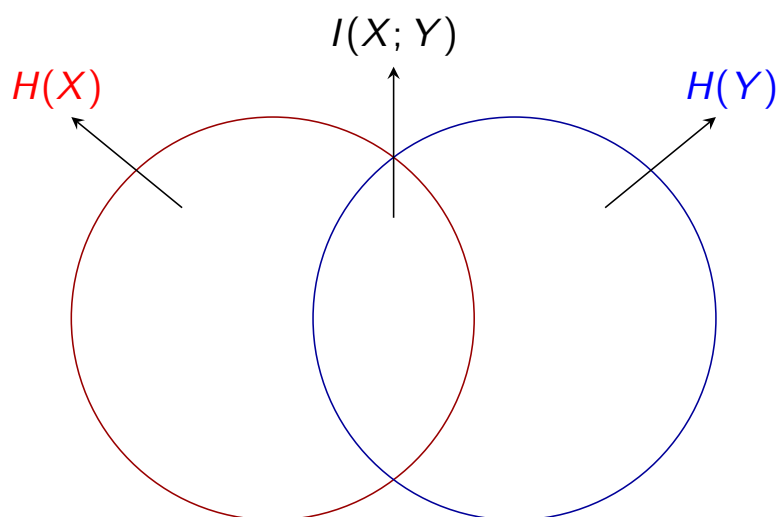
*Proof* : From the chain rule of entropy,

$$H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y).$$

In the definition of  $I(X; Y)$ , use  $H(X|Y) = H(X, Y) - H(Y)$ . □

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## Venn Diagram



The two circles together represent  $H(X, Y)$

## Questions

1. What is  $I(X; Y)$  when  $X$  and  $Y$  are independent?
2. What is  $I(X; Y)$  when  $Y = X$ ?
3. What is  $I(X; Y)$  when  $Y = f(X)$ ?

## Example

$X$  is the event that tomorrow is cloudy;  $Y$  is the event that it will rain tomorrow. Joint pmf  $P_{XY}$ :

|            | Rain   | No Rain |
|------------|--------|---------|
| Cloudy     | $3/8$  | $3/8$   |
| Not cloudy | $1/16$ | $3/16$  |

In Handout 1, we calculated:

$$H(X, Y) = 1.764, \quad H(X) = 0.811, \quad H(Y|X) = 0.953$$

To compute  $I(X; Y)$ , we need to compute  $H(Y)$  (or  $H(X|Y)$ ).

$$P(Y = \text{rainy}) = \frac{3}{8} + \frac{1}{16} = \frac{7}{16}, \quad P(Y = \text{not rainy}) = \frac{9}{16}$$

$$H(Y) = 0.989$$

$$I(X; Y) = H(Y) - H(Y|X) = 0.036$$

Verify that you get the same answer by computing  $H(X|Y)$  and using  $I(X; Y) = H(X) - H(X|Y)$ .

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### Property 2 of Mutual Information

$$I(X; Y) = D(P_{XY} \| P_X P_Y)$$

*“The relative entropy between the joint pmf and the product of the marginals”*

*Proof.*

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= - \sum_x P_X(x) \log P_X(x) + \sum_{x,y} P_{XY}(x, y) \log P_{X|Y}(x|y) \\ &= \sum_{x,y} P_{XY}(x, y) \log \frac{P_{X|Y}(x|y)}{P_X(x)} \\ &= \sum_{x,y} P_{XY}(x, y) \log \frac{P_{X|Y}(x|y) P_Y(y)}{P_X(x) P_Y(y)} \\ &= D(P_{XY} \| P_X P_Y). \end{aligned}$$

□

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### Property 3

$$I(X; Y) \geq 0$$

*Proof:* Follows from Property 2 because  $D(P\|Q) \geq 0$  for any pair of pmfs  $P, Q$ .

**Implication:**

$$H(X|Y) \leq H(X), \quad H(Y|X) \leq H(Y)$$

“Knowing another random variable  $Y$  can only reduce the average uncertainty in  $X$ ”

**Preview:**

- ▶ Let  $X$  be the input to a communication channel, and  $Y$  the output.
- ▶ We will show that  $I(X; Y)$  is key to understanding of how much information can be transmitted over the channel.

*“Reduction in the uncertainty of the channel input  $X$  when you observe the output  $Y$ ”*

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## Conditional Mutual Information and Chain Rule

Given  $X, Y, Z$  jointly distributed according to  $P_{XYZ}$ , the *conditional* mutual information  $I(X; Y|Z)$  is defined as

$$I(X; Y|Z) := H(X|Z) - H(X|Y, Z).$$

Mutual Information, like entropy, also obeys a chain rule:

For any sequence of random variables  $X_1, X_2, \dots, X_n$  jointly distributed with  $Y$ ,

$$I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_{i-1}, X_{i-2}, \dots, X_1),$$

where for each  $i$

$$\begin{aligned} & I(X_i; Y | X_{i-1}, X_{i-2}, \dots, X_1) \\ &= H(X_i | X_{i-1}, X_{i-2}, \dots, X_1) - H(X_i | Y, X_{i-1}, X_{i-2}, \dots, X_1). \end{aligned}$$

You will prove the chain rule for mutual information in the examples paper.

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